INCH-POUND
MIL-PRF-3098/140H
28 July 2011
SUPERSEDING
MIL-PRF-3098/140G
27 July 2005

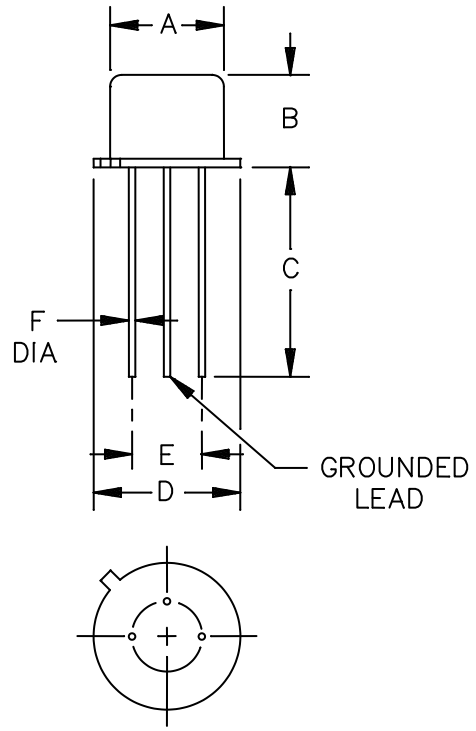
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR159/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 8 MHz to 20 MHz; fundamental; noncontrolled; series resonance.



Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.365	---	9.27
B	---	.265	---	6.73
C	.600	---	15.24	---
D	---	.420	---	10.67
E	.195	.205	4.95	5.21
F	.0175	.0205	0.445	0.521

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The crystal unit shall not be trimmed after sealing.
4. Marking shall be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR159/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 8 MHz to 20 MHz, inclusive.

Frequency tolerance : ± 50 ppm.

Equivalent resistance: See table I.

TABLE I Equivalent resistance .

Frequency range, inclusive. MHz	Maximum resistance Ohms
8.0 to 10.0	40
10.0+ to 12.0	30
12.0+ to 15.0	25
15.0+ to 20.0	20

Mode of oscillation: Fundamental.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Capacitance, shunt: 7.0 pF.

Shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 201, MIL-STD-202.

Frequency change permitted: ±10 ppm.

Equivalent resistance change permitted: ±15 percent.

Temperature cycling:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

Part or Identifying Number (PIN): For Product Level B, PIN shall be CR159/U or CR159B/U (followed by specified frequency). For Product Level S, PIN shall be CR159S/U (followed by specified frequency, see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

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(Project 5955-2011-015)

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